Gauge flow: Jacobian domains, Gaussian locality and unstable backgrounds

Maintained result. A specified background-covariant linear flow has an absolute Gaussian kernel-tail bound under smooth bounded coefficients and an integrable full vector curvature norm. The unrestricted Jacobian of Lüscher's gauge-modified flow has additional linear terms. Neither its identification with that auxiliary propagator nor a Hamiltonian relative truncation bound is proved by the Gaussian estimate.

A constant chromomagnetic background gives a complementary counterexample to universal contraction: the full aligned mode grows at $gB - k_\parallel^2$, including the Landau cost. It does not saturate the curvature-only majorant e^{2tgB} or give a discarded-tail lower bound. A free-field variance benchmark does not supply the necessary interacting supremum or probability estimate. These statements incorporate the earlier gauge, factor and sharpness corrections where their assumptions enter; source labels and written proofs follow.

1. The linearized flow and its gauge convention

Take Lüscher's gauge-modified continuum flow

$$\partial_s B_\mu = D_\nu G_{\nu\mu} + D_\mu q, \qquad q = \partial_\nu B_\nu.$$

The definition is from arXiv:1006.4518v3, equations (1.1)–(1.2) and (2.2), passage level via the local companion. Repeated spatial indices are summed. Write $u = \delta B$.

Proposition 1 (exact variation and auxiliary gauge choice). The Jacobian of this gauge-modified flow obeys

$$\partial_s u_\mu = D^2 u_\mu + 2[G_{\mu\nu}, u_\nu] + D_\mu(\partial_\nu u_\nu - D_\nu u_\nu) + [u_\mu, q].$$

The ungauged flow instead has linearization

$$\partial_s u_\mu = D^2 u_\mu - D_\mu D_\nu u_\nu + 2[G_{\mu\nu}, u_\nu].$$

Adding the **linear background-covariant gauge term** $D_\mu D_\nu u_\nu$ to the latter defines the auxiliary evolution used in Sections 2–4. Identifying it with another flow Jacobian requires an additional gauge comparison, not supplied here.

Proof. Since $\delta D_\nu = [u_\nu, \cdot]$ and $\delta G_{\nu\mu} = D_\nu u_\mu - D_\mu u_\nu$,

$$\delta(D_\nu G_{\nu\mu}) = D^2 u_\mu - D_\nu D_\mu u_\nu + [u_\nu, G_{\nu\mu}] = D^2 u_\mu - D_\mu D_\nu u_\nu + 2[G_{\mu\nu}, u_\nu].$$

Here $[D_\nu, D_\mu]u_\nu = [G_{\nu\mu}, u_\nu]$. The gauge term has the exact variation

$$\delta(D_\mu q) = D_\mu \partial_\nu u_\nu + [u_\mu, q],$$

not $D_\mu D_\nu u_\nu$. Adding these formulas proves the first identity; omitting the gauge term or adding the specified linear one proves the other assertions. □

Thus the original cancellation "to the same order" was unjustified: the omitted terms are themselves linear in u. The propositions below are about J, not about the full $D\Phi_t$ in Lüscher's convention. At $B = 0$ both equations reduce to the free heat equation. At a stationary background, background-transverse modes of the ungauged linearization also agree with the auxiliary generator; no general preservation of that transverse condition is asserted.

2. The Gaussian bound

Equip the compact Lie algebra with an invariant positive inner product, and its vector fields with $|u|^2 = \sum_\mu |u_\mu|^2$. The connection is metric-compatible because $\mathrm{ad}(B_\mu)$ is skew-adjoint. Assume $B(s,x)$ is smooth on $[0,t]\times\mathbb{R}^3$, with bounded coefficients and spatial derivatives sufficient for the covariant parabolic propagators and kernels below, and assume $\Gamma \in L^1([0,t])$. These are hypotheses, not estimates derived from action monotonicity.

Proposition 2. Under these assumptions the auxiliary propagator satisfies

$$\|J(t,0;x,y)\|_{\mathrm{op}} \le e^{A_t} K_t^{\mathrm{free}}(x-y), \qquad K_t^{\mathrm{free}}(z) = (4\pi t)^{-3/2} e^{-|z|^2/(4t)}.$$

In particular, if $\Gamma(s) \le \Gamma_*$ then $A_t \le 2t\Gamma_*$.

Proof. First consider $\partial_s v = D(s)^2 v$. Metric compatibility gives

$$(\partial_s - \Delta)|v|^2 = -2\sum_\alpha |D_\alpha v|^2.$$

Apply the chain rule to $h_\epsilon = (|v|^2+\epsilon^2)^{1/2}$ and use $|\partial_\alpha |v|^2| \le 2|v|\,|D_\alpha v|$. It follows that $(\partial_s - \Delta)h_\epsilon \le 0$. Scalar comparison followed by $\epsilon \downarrow 0$ yields, for the time-dependent covariant propagator $U(t,r)$,

$$|U(t,r)f|(x) \le (e^{(t-r)\Delta}|f|)(x).$$

With the assumed kernels this also gives $\|U(t,r;x,y)\|_{\mathrm{op}} \le K_{t-r}^{\mathrm{free}}(x-y)$, by testing localized initial data. This proves the needed Kato domination here; no bibliographic sharpness claim is used.

Set $M(s) = 2\mathcal{C}(s)$. Duhamel uses the two-time propagator,

$$J(t,0) = U(t,0) + \int_0^t U(t,s)M(s)J(s,0)\,ds.$$

In the term with n insertions, each multiplication factor has norm at most $2\Gamma(s_i)$ and scalar heat kernels convolve to $K_t^{\mathrm{free}}(x-y)$. The ordered time integral is bounded by $A_t^n/n!$. Summing this absolutely bounded series proves the result. □

Norm convention. If instead $g_* = \sup_{s,x,\mu,\nu} \|\,\mathrm{ad}(G_{\mu\nu}(s,x))\|_{\mathrm{op}}$, then $\Gamma_* \le 2g_*$ in three dimensions, or $\Gamma_* \le (d-1)g_*$ in d dimensions. Indeed

$$|(\mathcal{C}u)_\mu| \le g_* \sum_{\nu\ne\mu} |u_\nu|,$$

and Cauchy–Schwarz, followed by summation over μ, gives $|\mathcal{C}u|^2 \le (d-1)^2 g_*^2 |u|^2$. The former unspecified "adjoint operator norm" cannot silently omit this index factor.

At $B=0$, $J = K_t^{\mathrm{free}} I$ and the pointwise bound is attained. This free-case equality does not establish optimality of the curvature exponent for nonzero backgrounds.

3. Absolute kernel tails, not Hamiltonian relative errors

Define J_R by cutting its kernel to $|x-y| \le R$, and set

$$T_R = J - J_R, \qquad \delta(R,t) = e^{A_t}\int_{|z|>R} K_t^{\mathrm{free}}(z)\,d^3z.$$

Proposition 2 bounds both row and column integrals of $\|T_R(x,y)\|_{\mathrm{op}}$ by $\delta(R,t)$. Consequently the Schur test (or the scalar convolution majorant) gives the **absolute** bound

$$\|T_R\|_{L^2\to L^2} \le \delta(R,t).$$

It is not normalized by $\|J\|$ or by a kinetic quadratic form.

Corollary 3 (kernel-tail budget). Put $q = R/(2\sqrt{t})$. Then

$$\delta(R,t) = e^{A_t}\left[\operatorname{erfc}(q) + \frac{2q}{\sqrt{\pi}}e^{-q^2}\right] \le e^{A_t - q^2}\left(1 + \frac{2q}{\sqrt{\pi}}\right).$$

Thus at $R = \kappa\sqrt{8t}$ a sufficient condition for $\|J - J_R\|_{2\to 2} \le \epsilon$ is

$$2\kappa^2 \ge A_t + \log\left(1 + \frac{2\sqrt{2}\kappa}{\sqrt{\pi}}\right) + \log(1/\epsilon), \qquad 0 < \epsilon < 1.$$

For every finite A_t this is achieved at sufficiently large κ. It is a sufficient condition only.

Proof. Radial integration and $r = 2\sqrt{t}\,v$ give

$$\int_{|z|>R} K_t^{\text{free}}(z)\, d^3z = \frac{4}{\sqrt{\pi}}\int_q^\infty v^2 e^{-v^2} dv = \operatorname{erfc}(q) + \frac{2q}{\sqrt{\pi}}e^{-q^2}.$$

For $q \ge 0$, $\operatorname{erfc}(q) \le e^{-q^2}$: their difference is zero at zero, has derivative $2e^{-q^2}(1/\sqrt{\pi} - q)$, and tends to zero at infinity. The stated inequality and logarithmic budget follow. □

At $\kappa = 1$, even the exact free discarded mass is $\operatorname{erfc}(\sqrt{2}) + (2\sqrt{2}/\sqrt{\pi})e^{-2}$. This is a fixed positive number, not a parameter tending to zero on small fields. A growing upper bound for A_t implies neither a large actual tail nor a lower bound on a truncation error.

The Hamiltonian conversion remains open. The kinetic metric in the flow-conjugation note, §§1–3 (full-read), involves quadratic Jacobian combinations and inverse metrics; the transformed operator also has density derivatives and lower-order terms. Already for bounded operators F and F_R, with $E = F - F_R$,

$$\|F^*F - F_R^*F_R\| \le 2\|F\|\,\|E\| + \|E\|^2.$$

This follows by expanding $F_R = F - E$. It is an absolute estimate; a relative comparison additionally needs a positive lower bound on the relevant metric or a bound on its inverse. The free continuum heat multiplier $e^{-t|k|^2}$ has no bounded inverse on L^2. The finite lattice has an invertible flow Jacobian, but uniform inverse, configuration-derivative and relative form estimates still need proof. The auxiliary J above is not identified with that lattice Jacobian. Thus the original Hamiltonian interpretation of δ remains a **heuristic motivation**, not a proved relative truncation theorem.

4. The sufficient small-field budget and its limits

Write $\ell = \sqrt{8t}$. If the block norm satisfies

$$\Gamma(s) \le \frac{\eta}{\ell^2} \quad (0 \le s \le t),$$

then $A_t \le \eta/4$ and Corollary 3 yields

$$\|J - J_R\|_{2\to 2} \le \left(1 + \frac{2\sqrt{2}\kappa}{\sqrt{\pi}}\right)\exp\left[\frac{\eta}{4} - 2\kappa^2\right].$$

The original 2η growth exponent in this substitution was incorrect. In component norm, the hypothesis $g_* \le \eta/\ell^2$ gives instead $A_t \le \eta/2$ in three dimensions by the index bound of Section 2.

regime or question	supported conclusion
bounded curvature budget A_t, sufficiently long range	absolute auxiliary kernel tail is below a prescribed tolerance
small η at one fixed flow radius	bounded tail, with no arbitrarily small accuracy implied
large A_t at fixed range	this upper bound deteriorates; actual tail size remains undetermined
larger range at finite A_t	the kernel majorant can be made small
conjugated Hamiltonian or iterated RG	relative form and iteration estimates remain open

This budget motivates comparison with small/large-field splitting in constructive renormalization, but neither forces that split nor controls all steps of the flow-conjugation route. The comparison to Balaban's series (Commun. Math. Phys. 122 (1989) 355, metadata level) is context, not a derivation of its hypotheses. No unproved decay of a "typical" interacting curvature is used here.

5. What action monotonicity controls

For smooth finite-action solutions of the ungauged gradient flow, with decay allowing integration by parts and the gradient normalization,

$$\frac{d}{ds}S(B_s) = -\|\nabla S(B_s)\|_2^2 \le 0.$$

The added gauge term is an infinitesimal gauge transformation and does not change this identity under the same boundary assumptions. Since S is proportional to $\int |G|^2$, it gives

$$\|G(t)\|_2^2 \le \|G(0)\|_2^2.$$

It supplies no L^∞ bound by itself, and Proposition 2 requires control over the **whole flow interval**, not just its endpoint.

Free or abelian smoothing, corrected. Around the zero connection J is the heat semigroup. In an abelian theory the curvature itself obeys the heat equation, so Cauchy–Schwarz and the Gaussian convolution identity give

$$\|e^{t\Delta} f\|_\infty \le \|K_t^{\mathrm{free}}\|_2 \, \|f\|_2 = (8\pi t)^{-3/4}\|f\|_2.$$

For abelian initial curvature this bounds $\|G(t)\|_\infty$ by $(8\pi t)^{-3/4}\|G(0)\|_2$ in the Euclidean component norm. For fixed nonzero initial norm its ratio to $1/t$ is proportional to $t^{1/4}\|G(0)\|_2$: it tends to zero for small t and grows for large t, the reverse of the former remark. Integrating this estimate over $0 < s < t$ also gives a finite budget proportional to $t^{1/4}\|G(0)\|_2$, using the finite-dimensional conversion to the block norm. This is an abelian estimate, not an interacting theorem. The coefficient $(4\pi t)^{-3/4}$ formerly displayed is a looser valid bound; the expression above is the exact heat-kernel L^2 norm.

Monotonicity alone does not supply the missing norm. A fixed L^2 curvature norm permits concentration in small regions with arbitrarily large height. For example rescale a smooth compactly supported abelian connection by $B_\lambda(x) = \lambda^{1/2}B(\lambda x)$; its curvature is $G_\lambda(x) = \lambda^{3/2}G(\lambda x)$, with constant L^2 norm and unbounded supremum as $\lambda \to \infty$. This disproves a pointwise estimate from the action bound alone, not parabolic estimates using additional evolution information.

The Hamiltonian formulation also has configuration probabilities, for example its ground-state density; see the transfer note, §§1–2 (full-read). Such measure estimates do not automatically give the signed operator comparisons needed for spectral stability. The distinction

emphasized in the operator-inequality note, §3 (full-read), is relevant here; this note does not assume its claimed Hamiltonian truncation estimate or conclude that every Hamiltonian route fails.

Instability of the full background generator

Let $\mathfrak{g}$ be a compact matrix Lie algebra with invariant inner product $\langle X, Y\rangle = -\operatorname{tr}(XY)$. Then $\operatorname{ad}(Z)$ is antisymmetric: $\langle [Z, X], Y\rangle = -\langle X, [Z, Y]\rangle$.

Proposition 1. At each point, on $\mathfrak{g}$-valued vectors with inner product $\sum_\mu \langle u_\mu, v_\mu\rangle$, the operator $(Mu)_\mu = 2[G_{\mu\nu}, u_\nu]$ is symmetric.

Proof.

$$\langle Mu, v\rangle = 2\sum_{\mu\nu}\langle [G_{\mu\nu}, u_\nu], v_\mu\rangle = -2\sum_{\mu\nu}\langle u_\nu, [G_{\mu\nu}, v_\mu]\rangle = 2\sum_{\mu\nu}\langle u_\nu, [G_{\nu\mu}, v_\mu]\rangle = \langle u, Mv\rangle.$$

The two antisymmetries cancel. □

A finite-dimensional symmetric operator of norm N has an eigenvalue of modulus N; symmetry does not determine its sign. Moreover, positive eigenvalues of M alone do not determine those of $D^2 + M$. For compactly supported fields, integration by parts gives

$$\langle u, Lu\rangle = -\sum_\alpha \|D_\alpha u\|_2^2 + \langle u, Mu\rangle.$$

The first term must be retained.

The preceding Duhamel bound uses $\Gamma = \frac{1}{2}\sup\|M\|$; its exponent is an upper bound only. The following example retains both terms of L.

The charged Landau mode

We work with the Yang–Mills flow linearized at a stationary background $\bar{A}$, in **background-covariant gauge**. This specifies the operator rather than assuming that every gauge-modified flow has the same Jacobian. With $D = \partial + [\bar{A}, \cdot]$ and $D_\nu G_{\nu\mu} = 0$, variation of the ungauged flow gives

$$\partial_s u_\mu = D^2 u_\mu - D_\mu D_\nu u_\nu + 2[G_{\mu\nu}, u_\nu].$$

Adding the linear background-gauge term $D_\mu D_\nu u_\nu$ gives L. Equivalently, on a mode with $D_\nu u_\nu = 0$, the ungauged linearization already agrees with L. This does not repair or identify the full Jacobian in every gauge convention of the gauge convention above.

Choose an $SU(2)$ colour sector with $[T^3, E_+] = iE_+$, and absorb the coupling into the connection, so $G_{12} = bT^3$, $b = gB > 0$. In symmetric gauge on the transverse plane,

$$\bar{A}_1 = -\frac{b}{2}x_2 T^3, \qquad \bar{A}_2 = \frac{b}{2}x_1 T^3.$$

On the E_+ component,

$$D_1 = \partial_1 - \frac{ibx_2}{2}, \qquad D_2 = \partial_2 + \frac{ibx_1}{2}.$$

Writing $Q = D_1 - iD_2$, integration by parts and $[D_1, D_2] = ib$ give $Q^*Q = -(D_1^2 + D_2^2) - b$ and $[Q, Q^*] = 2b$. Thus the oscillator ladder has transverse magnetic-Laplacian energies $(2n+1)b$, $n = 0, 1, \ldots$. To fix the factor without relying on a bibliographic convention, its lowest mode can be checked directly:

$$f(x_1, x_2) = e^{-b(x_1^2+x_2^2)/4}, \qquad (D_1 - iD_2)f = 0, \qquad (D_1^2 + D_2^2)f = -bf.$$

On the two transverse vector components,

$$M = \begin{pmatrix} 0 & 2ib \\ -2ib & 0 \end{pmatrix}, \qquad M\begin{pmatrix} f \\ -if \end{pmatrix} = 2b\begin{pmatrix} f \\ -if \end{pmatrix}.$$

The charged complex notation represents real Lie-algebra fields by adding the conjugate charge sector. The vector

$$u = (f, -if, 0, \ldots)e^{ik_\parallel \cdot x_\parallel}$$

is background-transverse, since $D_1 f - iD_2 f = 0$. Consequently

$$D^2 u = -(b + k_\parallel^2)u, \qquad Lu = (b - k_\parallel^2)u.$$

This supplies both the positive eigenvalue of M and the Laplacian cost on the same vector. The two transverse polarizations at general Landau level have the spectral values

$$\omega^2 = k_\parallel^2 + (2n+1)b \mp 2b$$

for $-L$. Here $k_\parallel^2$ is the sum of the squared momenta in directions outside the magnetic plane: one for spatial three-dimensional flow, two for four-dimensional Euclidean flow. The unstable branch is

$$\omega^2 = k_\parallel^2 - b < 0 \quad \text{when } k_\parallel^2 < b,$$

and its gradient-flow growth is

$$e^{-\omega^2 s} = e^{(b - k_\parallel^2)s}.$$

At $k_\parallel = 0$ the rate is b, half the exponent coefficient $2b$ in the Duhamel majorant. That example therefore **does not saturate** the majorant. Whether a better universal bound holds for other backgrounds is not settled by this calculation.

On the whole space the longitudinal plane waves are generalized modes; wave packets supported in $k_\parallel^2 < b$ give growing L^2 perturbations, with rates approaching b. Periodic longitudinal boundary conditions admit the zero-momentum mode. The constant field on the transverse plane has infinite total action; the calculation is a stationary-background spectral model, not a finite-action four-dimensional construction or a statement for arbitrary finite boxes.

This is the instability historically associated with Nielsen and Olesen, "An unstable Yang–Mills field mode," Nucl. Phys. B144 (1978) 376–396 (abstract read; the original spectral passage was not read here). The numerical factors used above are supported by the displayed calculation, not by that abstract or a normalization left implicit.

Action descent and growing perturbations

For the unmodified gradient flow with its gradient-metric normalization,

$$\frac{d}{ds} S(A_s) = -\|\nabla S(A_s)\|_2^2 \le 0.$$

A constant abelian curvature has $D_\nu G_{\nu\mu} = 0$ and is stationary. Its transverse unstable modes give negative Hessian directions, $\operatorname{Hess} S = -L$ on the background-transverse sector in the normalization of Section 2. Nearby trajectories can separate at rate $b - k_\parallel^2$ while descending the action. For the infinite-action background these are statements about the local equation and quadratic variation on admissible perturbations, rather than differentiation of an infinite total action.

The original comparison with the valley note is also corrected: constant commuting connections in its flat holonomy valley have $G = 0$. They are not the nonzero constant chromomagnetic background considered here. The valley's zero-point potential is a different calculation and gives no justification for this saddle or its growth rate.

Free-field size is a benchmark, not a supremum bound

From the free-field calculation §5c, the flowed magnetic two-point function at coincident points is

$$\langle b_{t,i}(0)b_{t,i}(0)\rangle = \int \frac{d^3k}{(2\pi)^3}\sum_i \tilde{G}_{ii}(k) = \int \frac{d^3k}{(2\pi)^3}\,\frac{g^2k}{2}\,e^{-2tk^2}\cdot 2 = \frac{g^2}{2\pi^2}\int_0^\infty k^3 e^{-2tk^2}dk,$$

and $\int_0^\infty k^3 e^{-\alpha k^2}dk = 1/(2\alpha^2)$ with $\alpha = 2t$ gives

$$\langle |b_t|^2\rangle = \frac{g^2}{2\pi^2}\cdot\frac{1}{8t^2} = \frac{g^2}{16\pi^2 t^2}, \qquad \langle |b_t|^2\rangle^{1/2} = \frac{g}{4\pi\, t}.$$

The flow time is the only scale, so the dimensionless combination is

$$t\,\langle |b_t|^2\rangle^{1/2} = \frac{g}{4\pi},$$

independent of t and of the lattice spacing. In the interacting theory at scale $\ell = \sqrt{8t}$ the same estimate with the running coupling $g(\ell)$ is the leading perturbative statement.

The displayed convention absorbs the quantum factor in g^2; writing $g^2 = \hbar g_{\rm cl}^2$ restores physical $\hbar$ explicitly. A single-point Gaussian variance is neither a bound on the field supremum over a continuum region nor on the whole-interval curvature block norm. Substitution of $g/(4\pi t)$ into a deterministic kernel hypothesis is only a model estimate. Even for a Gaussian vector field, its component variance, radial tail, spatial correlation and continuum supremum require their own constants; the old scalar exponent $8\pi^2\eta^2/g^2$ and $V/(8t)^{3/2}$ independent-cell count did not prove such a theorem. Replacing g by a running coupling is a perturbative suggestion, not a nonperturbative interacting probability bound.

At $R = \kappa\sqrt{8t}$, the earlier schematic substitution gave $\exp[g/(2\pi) - 2\kappa^2]$ and $\kappa^2 \geq g/(4\pi) + 1/2$ for tolerance e^{-1}. The exact tail above additionally has a polynomial prefactor and uses a supremum over all flow times. These numbers therefore remain illustrative scale bookkeeping, not a Hamiltonian truncation guarantee at any coupling. Hamiltonian ground states have configuration probabilities; their transfer identity does not turn an absolute auxiliary kernel bound into the required signed operator comparison.

Consequence for STATE

Keep exact gauge variation, auxiliary kernel locality, background instability, free-field statistics and Hamiltonian form comparison as separate steps. At $\ell = \sqrt{8t}$ and $\Gamma \leq \eta/\ell^2$, the growth budget is $\eta/4$; a sufficiently long range controls this kernel upper bound for any finite budget. No sharp cutoff crossover, normalized large-field tail, large-field exclusion theorem or continuum mass gap follows. The finite-lattice conjugation and signed operator comparison retain their independent domains and missing estimates.